

software to design and print, and materials used in printing.

According to the survey participants, 3D printers based on fused deposition modelling (FDM) technology dominate the Indian market (Fig. 4). This type of printer has become popular due to its low cost and easy availability. However, stereo lithography (SLA) and selective laser sintering (SLS) based printers are also gaining prominence.

The printers need to be improved to cater to the future needs beyond prototyping. The views of survey participants on the improvement needs can be broadly categorised as:

- Performance improvements with respect to speed, resolution, autonomous operation, ease of use, reliability and repeatability
 - Capability to incorporate multiple types of materials and the ability to mix materials while printing a single object
 - Ability to print fully functional and active systems that contain many modules such as embedded sensors, batteries, electronics, micro-electromechanical systems (MEMS), etc
- The availability of 3D printers for direct short-run manufacturing with metal alloys like titanium, cobalt chrome, Inconel and other materials opens up opportunities in aerospace, defence and medical sectors. The possibility to print with a wider variety of materials at higher speed and accuracy is something that will make this sector more dynamic.

According to survey participants, open source SLS technology and new techniques for metal 3D printing are also driving growth. Other interesting technological trends include:

- Availability of industrial plastic materials for making direct-use plastic components
- Distributed/personal printing of

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- household requirements
- Use of 3D printing in wearable technologies
- Direct CAD-to-gold 3D printing for customised jewellery manufacturing

Obstacles to growth

Presently, low market awareness, cost constraints and lower domestic production hinder large-scale adoption of 3D printing technologies in the country. Specifically, awareness about this technology is yet to reach SMEs, small scale industries and individuals, which may constitute a major part of the consumers.

According to the survey participants, the challenges facing this industry include:

- Inefficient supply chain for raw materials and components required for 3D printer manufacturing
- No government recognition for 3D printing as an industry
- Heavy import duties for 3D printers
- Limited availability of experience zones, service centres and franchises for Indian 3D printer manufacturers
- Performance limitations of existing low-cost printers

- High cost of advanced SLA, SLS, polyjet and multi-jet 3D printers make these out of reach for small manufacturers
- Market infiltration by poor-quality products and inefficient industry players

Moreover, the 3D printer market faces a major challenge from cheaper CNC milling machines (used for modelling) due to the cost sensitivity issue in India. This results in reluctance among consumers to purchase 3D printers.

The road ahead

The Indian 3D printing industry can witness healthy growth in the near future, considering the expected price decline and increasing market awareness, coupled with major transformations such as easier design capabilities, compatibility for mass production and manufacture of large-format objects. The Indian 3D printing industry is not yet at a stage to create the 'factory at every home' atmosphere, but with the Indian government's initiatives to boost the domestic manufacturing sector, this industry has the potential to facilitate the next industrial revolution in the country. **EFY**